Point of Care: Modernizing Bedside Charting for Rural Hospitals with HL7v2 Interoperability

Data Bridge

Software Requirements Specification

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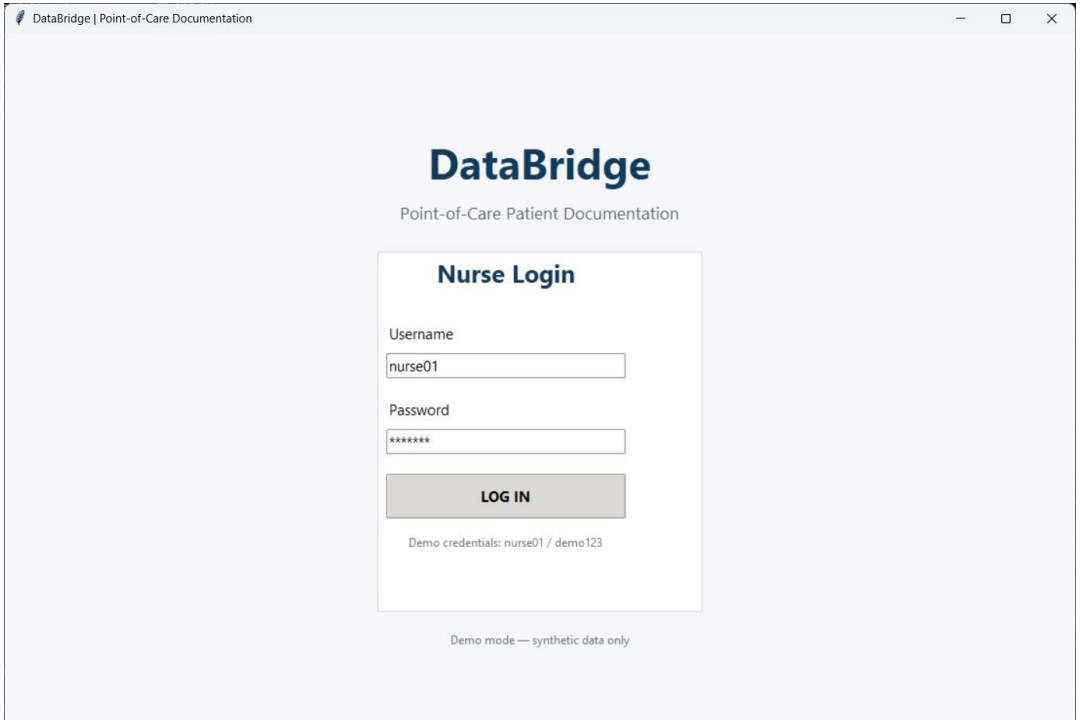
Introduction

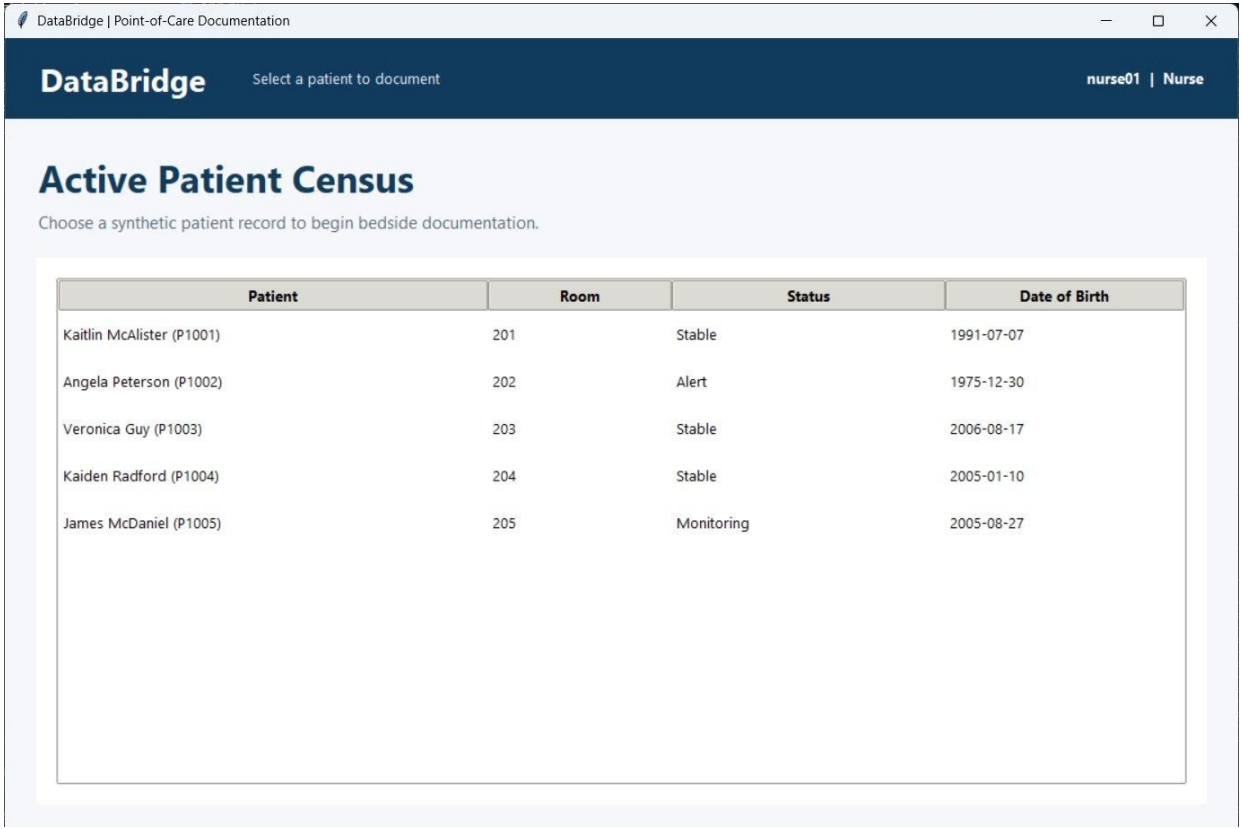
- Purpose: Defines the requirements for DataBridge, a browser-based bedside documentation system for rural hospitals on legacy MEDITECH platforms.
- Audience: Development team, Project Mentor and Course Evaluators
- Scope:Nurses enter vitals and notes on a tablet instead of paper. Each entry is converted into an HL7v2 message and sent to a simulated EHR receiver. Goal is to reduce delays and errors from double documentation.

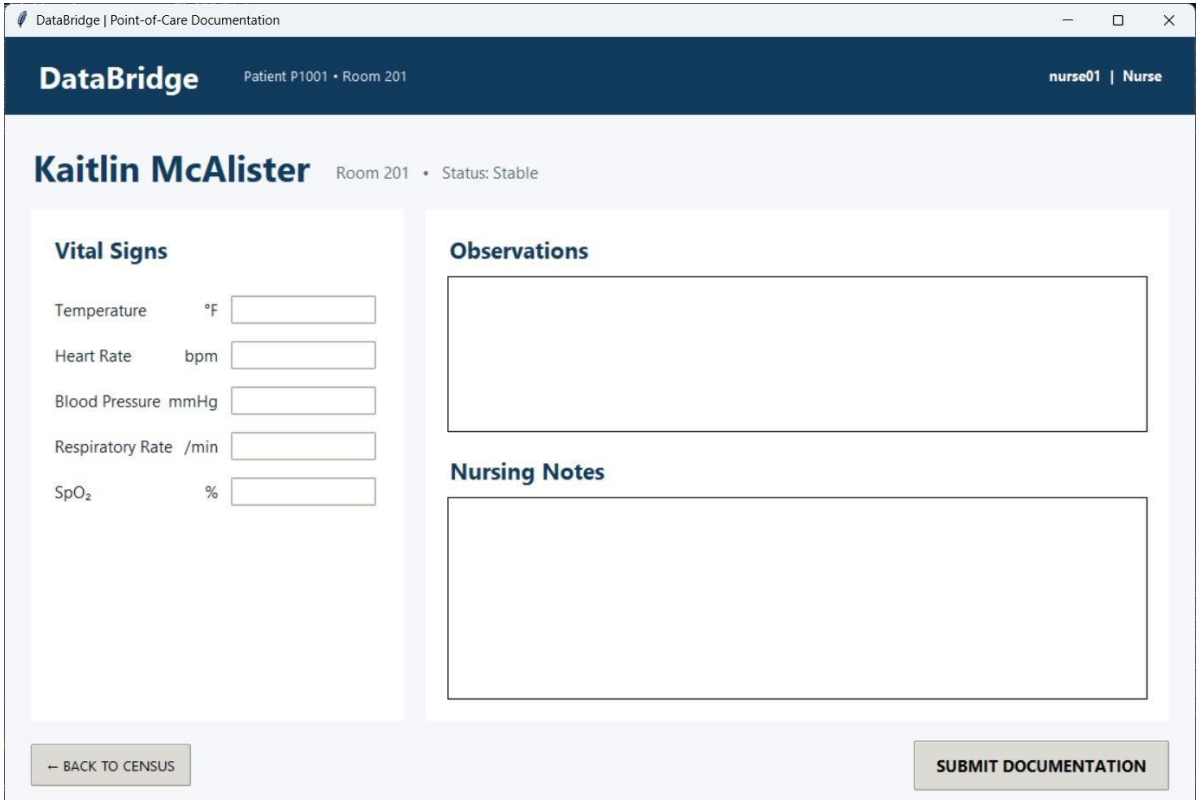
Overall Description

- Product Perspective: A lightweight layer between the nurse and the hospital's existing EHR. Made up of a table frontend, a backed that validates and converts data and a receiver.
- User Classes:
  - Nurse: Primary user, documents vitals for assigned patients
  - Charge Nurse: Views the full unit census
  - Administrator: Manages users and reviews audit logs.
- Operating Environment: Web browser on iPad or desktop. Backend in Python with FastAPI
- Constraints: No live hospital access, synthetic data only, HL7v2 used because legacy MEDITECH uses it.
- Assumptions: Relies on open-source HL7v2 libraries, Synthea for synthetic data, and free tier hosting.
- User Documentation: Short user guide for nurses and an administrator guide.

External Interface Requirements

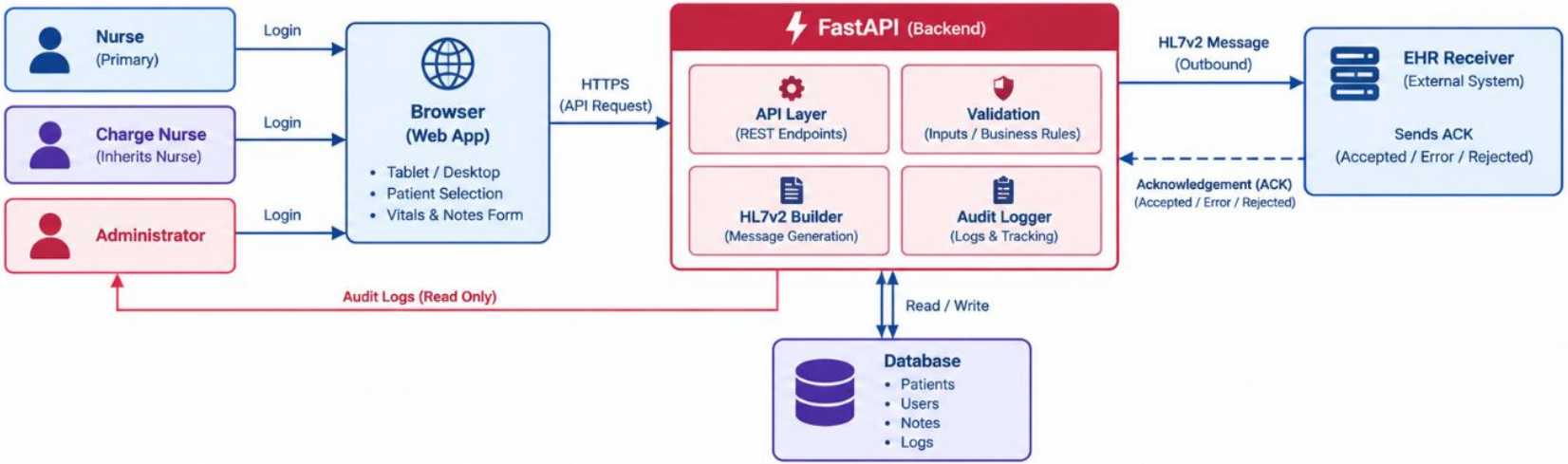



DataBridge | Point-of-Care Documentation
DataBridge
Select a patient to document
nurse01 | Nurse
Active Patient Census
Choose a synthetic patient record to begin bedside documentation.
Patient
Room
Status
Date of Birth
Kaitlin McAlister (P1001)
201
Stable
1991-07-07
Angela Peterson (P1002)
202
Alert
1975-12-30
Veronica Guy (P1003)
203
Stable
2006-08-17
Kaiden Radford (P1004)
204
Stable
2005-01-10
James McDaniel (P1005)
205
Monitoring
2005-08-27

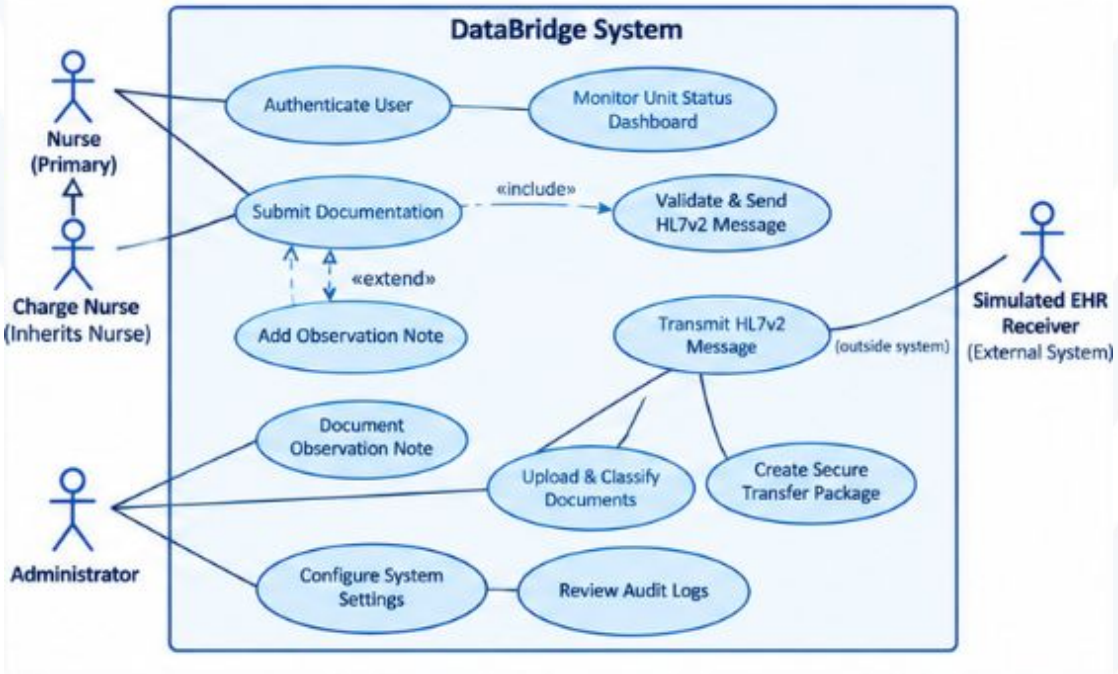

DataBridge | Point-of-Care Documentation
DataBridge
Patient P1001 • Room 201
nurse01 | Nurse
Kaitlin McAlister
Room 201 • Status: Stable
Vital Signs
Temperature °F
Heart Rate bpm
Blood Pressure mmHg
Respiratory Rate /min
SpO₂ %
Observations
Nursing Notes
← BACK TO CENSUS
SUBMIT DOCUMENTATION

System Features

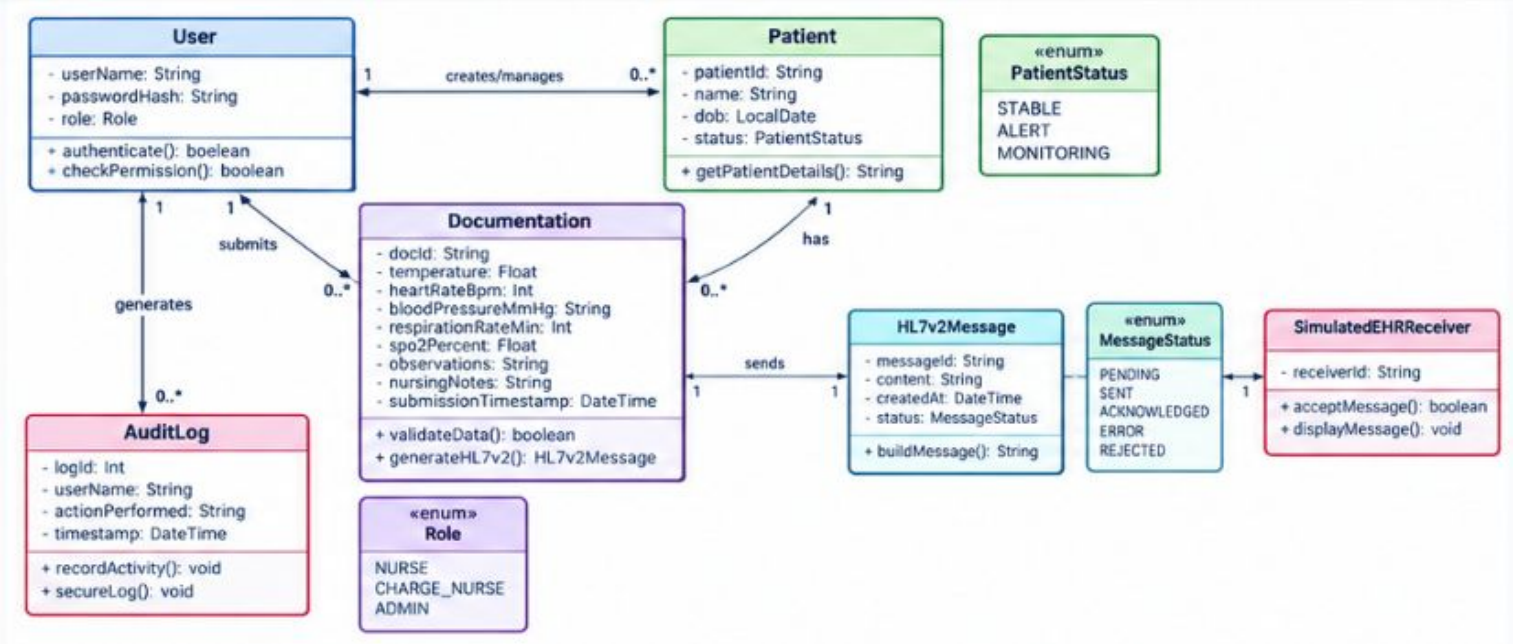
- High Priority:
  - User Authentication
  - Role-Based Access Control
  - Bedside Documentation
  - Backend Validation and Processing
  - HL7v2 Message Generation
  - Simulated EHR Receiver
- Medium Priority:
  - Audit Logging
  - Document Upload, Classification, and OCR
  - Secure Document Transfer


Nurse
(Primary)
Charge Nurse
(Inherits Nurse)
Administrator
Login
Login
Login
Browser
(Web App)
Tablet / Desktop
Patient Selection
Vitals & Notes Form
HTTPS
(API Request)
FastAPI (Backend)
API Layer
(REST Endpoints)
Validation
(Inputs / Business Rules)
HL7v2 Builder
(Message Generation)
Audit Logger
(Logs & Tracking)
HL7v2 Message
(Outbound)
Acknowledgement (ACK)
(Accepted / Error / Rejected)
EHR Receiver
(External System)
Sends ACK
(Accepted / Error / Rejected)
Audit Logs (Read Only)
Read / Write
Database
Patients
Users
Notes
Logs

Use-Case Main Structure



UML Diagram


User
- userName: String
- passwordHash: String
- role: Role
+ authenticate(): boolean
+ checkPermission(): boolean
1
creates/manages
0..*
Patient
- patientId: String
- name: String
- dob: LocalDate
- status: PatientStatus
+ getPatientDetails(): String
«enum»
PatientStatus
STABLE
ALERT
MONITORING
1
1
submits
0..*
1
has
0..*
generates
0..*
Documentation
- docId: String
- temperature: Float
- heartRateBpm: Int
- bloodPressureMmHg: String
- respirationRateMin: Int
- spo2Percent: Float
- observations: String
- nursingNotes: String
- submissionTimestamp: DateTime
+ validateData(): boolean
+ generateHL7v2(): HL7v2Message
sends
1
1
HL7v2Message
- messageId: String
- content: String
- createdAt: DateTime
- status: MessageStatus
+ buildMessage(): String
«enum»
MessageStatus
PENDING
SENT
ACKNOWLEDGED
ERROR
REJECTED
1
SimulatedEHRReceiver
- receiverId: String
+ acceptMessage(): boolean
+ displayMessage(): void
AuditLog
- logId: Int
- userName: String
- actionPerformed: String
- timestamp: DateTime
+ recordActivity(): void
+ secureLog(): void
«enum»
Role
NURSE
CHARGE_NURSE
ADMIN

Other Nonfunctional Requirements

- Performance: Responsive interface: feedback after submission, full workflow runs without manual steps.
- Safety: No diagnoses or treatment recommendations. Invalid data is blocked. Synthetic Data only.
- Security: Password plus MFA, role-based access with least privileged, encryption in transit and at rest, audit logging, expiring securing sharing links. HIPAA-aligned but not HIPAA-certified.
- Quality Attributes: Easy to use on a tablet, reliable error handling, HL7v2 interoperability, separated components, testable with synthetic data.
- Business Rules: Users must log in, every user has a role, users only access what their role allows, documentation is tied to the user and patient, simulated receiver used instead of live EHR

Thank you for listening!

Github Repo:

